Sattwik Banerjee

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Summary

Early-career data science professional with hands-on experience in machine learning model development and data analysis. Developed Bayesian Belief Network models to predict pipe corrosion and reinforcement learning frameworks to optimize maintenance scheduling. Built a vector database for semantic search and collaborated on large-scale genomic analyses, seeking to contribute analytical and computational expertise as a machine learning intern.

EDUCATION

University of California, Los Angeles **Jun 2027**
Bachelors of Science, Data Theory and Data Science Engineering

- **GPA:** 3.7
- **Coursework:** Probability and Statistical Theory, Mathematical Methods of Data Theory, Machine Learning, Linear Algebra and Applications, Data Science Fundamentals, Data Structures and Algorithms, Statistical Analysis with R/Python, SQL, C++

Pasadena City College, Pasadena **Jun 2024**
Associates of Art, Engineering and Technology

- **GPA:** 3.8
- **Coursework:** Probability and Statistical Theory, Mathematical Methods of Data Theory, Machine Learning, Linear Algebra and Applications, Data Science Fundamentals, Data Structures and Algorithms, Statistical Analysis with R/Python, SQL, C++

SKILLS

- **Language**: Python, R, C++, SQL, HTML/CSS
- **Developer Tools**: VS Code, PyCharm, Git, Pandas, Matplotlib, Plotly, Numpy, RStudio, PyTorch, TensorFlow, Machine Learning Frameworks, Model Deployment
- **Interests**: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Data Analysis, Big Data, Medical Applications of AI

WORK EXPERIENCE

The B. John Garrick Institute for Risk Sciences UCLA **Sep 2025 - Present**
Undergraduate Researcher *Los Angeles, CA*

- Developed Bayesian Belief Network (BBN) models to characterize three distinct types of pipe corrosion, integrating environmental and material variables for probabilistic fault prediction using machine learning frameworks such as PyTorch and TensorFlow.
- Designed a reinforcement learning framework that uses BBN-derived corrosion probabilities to optimize maintenance scheduling and determine when pipeline repair should occur, incorporating model deployment best practices.

CloudSDS Inc. **Jun 2025 - Aug 2025**
Data Science Intern *Los Angeles, CA*

- Developed a vector database using Python to enable semantic search across manufacturer and product data, allowing for natural language queries.

Dr. Paul Boutros Lab, Cancer Data Sciences UCLA **Aug 2024 - Sep 2025**
Undergraduate Researcher *Los Angeles, CA*

- Utilized advanced statistical methods and machine learning algorithms to analyze large-scale medical datasets in R, optimizing workflows through High Performance Computing (HPC).
- Collaborated with a team of 5+ researchers to analyze genomic data from 1,000+ cancer patients, identifying biomarkers linked to 2.5x higher recurrence risk.

PROJECTS

Protein Secondary Structure Prediction

- Fine-tuned an ESM-2 protein language model to predict 9-class secondary structure from amino-acid sequences, building a custom PyTorch training pipeline and achieving 82% validation accuracy.

NBA RAG Search Engine

- Built a retrieval-augmented generation system enabling natural-language queries over NBA player, team, and game datasets.
- Integrated Llama models via LangChain for reasoning and natural-language synthesis.

Heart Disease Detection Using ML

- Developed an Artificial Neural Network to determine the likelihood a patient has heart disease depending on numerous features, achieving a Root Mean Squared Error of 0.298 implying high model accuracy.